

KETRO SITHOLE.

Bsc Computer Sciences Graduate — University of Pretoria.

Bsc Honours Computer Science – Artificial Intelligence & Machine Learning

Energetic, Inquisitive, Intelligent, and Resourceful Problem Solver.

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Professional Profile :

Published Author (Machine Learning) – *“An Automatically Collected, Prepared, and Benchmarked Dataset for South African Sign Language.”*

Lead Data Engineer with 3+ years of experience delivering scalable, end-to-end data solutions across banking, financial services, academia, and the UK BPO sector. I have worked with FNB, Nedbank, and SBV, specialising in data warehousing, analytics engineering, and cloud-based data platforms, including SSIS-based ETL solutions.

As an Assistant Lecturer, I teach Data Science, Statistics, Data Warehousing, and Applied Data Science. I also provide instruction in GitHub for version control, Ruby on Rails, and Golang, emphasizing industry-aligned best practices, clean architecture, and real-world problem-solving. My teaching approach equips students to tackle data-driven decision-making and complex analytical challenges with confidence.

I am highly skilled in Python, SQL, Apache Spark, Microsoft Azure, Databricks, Snowflake, Docker, Power BI, SSIS, and GitHub, with working knowledge of Ruby on Rails and Golang for backend and data-driven application development.

Recognised as one of South Africa's Top 15 Young Geeks (2023) and a seven-time hackathon winner, I am committed to mentorship, technical leadership, and community-driven technology education. I graduated with First Place for my coding capstone project and am a highly adaptable, intellectually curious engineer who consistently seeks to expand expertise in data engineering, artificial intelligence, and large-scale analytical systems.

Objectives:

I am eager to join high-performing engineering teams where I can leverage my expertise as a Data Engineer or BI Developer to drive meaningful social impact, including combating cybercrime, strengthening secure and resilient data systems, and bridging the digital skills gap. I am intentionally planning my growth toward a CTO or Principal Engineering leadership role over the next few years, combining deep technical depth with strategic thinking, mentorship, and systems-level impact. I am committed to building scalable, automated, and secure data solutions grounded in ethical data practices, collaboration, and innovation, supporting sustainable growth across South Africa and the global digital economy.

Academic Qualifications & Professional Certifications : [Completed Matric](#) → [Bsc Degree](#) → [Bsc Honours](#) → [Microsoft certifications](#)

BSc Computer Science – University of Pretoria : *Completed.*

- **Majors:** Computer Science, Economics, Business Management, Statistics, Data Science
- **Electives:** Competitive Intelligence, Digital Repositories, Information Ethics, Information Organization

Bsc Honours Computer Science – Artificial Intelligence & Machine Learning :

- **Research Focus:** Predictive Sciences and Financial Engineering.
- **Modules :** Microservices Architectures, Big Data Analytics Techniques, and Time Series Analysis and Forecasting

Certifications :

- **Microsoft Certified:** Azure Cloud Fundamentals (AZ-900) -Passed
- **Microsoft Certified:** Azure Data Fundamentals (DP-900) - Passed

Work Experience :

August 2025 – Present | 6-Months | Lead Data Engineer | Sigma Connected Group | United Kingdom-Remote | Fulltime

I'm a Senior Data Engineer specializing in credit data, enabling the business to track outstanding balances, manage credit risk, and support effective collections and customer outreach. I build reliable data pipelines using SSIS, Azure DevOps, Python, SQL, Golang, and Azure, working with microservices and enterprise database tools to deliver accurate, timely, and trusted data for business decision-making.

1. **Programming :** Golang, Python, SQL, NoSQL, FastAPI, Ruby , Powershell ,Bat
2. **Data Engineering:** SSIS (ETL/ELT), Data Warehousing, Streaming, Pipeline Optimization, Process Gold
3. **Software Engineering:** Microservices, RESTful APIs, System Design, Cloud-Native Architectures
4. **DevOps & DataOps:** CI/CD, Docker, Kubernetes, High Availability, RPA Automation using UiPath

Jan 2024 -July 2025 | Business Intelligence Developer and Data Engineer | Expeditus | Full-Time

I worked as an **on-site Business Intelligence Developer and Data Engineer** supporting **FNB, Nedbank, and SBV**, where I designed and implemented scalable data lakes and data warehouses to enable enterprise analytics. I enforced data governance, security, and regulatory compliance, optimized SQL performance, and built large-scale ETL/ELT pipelines for structured and unstructured data using Spark, Hadoop, and cloud-native architectures on Microsoft Azure. I also led and mentored junior engineers, reviewed data pipelines and architectures, and collaborated with cross-functional teams to align technical delivery with business objectives.

- **Programming:** Python, SQL, NoSQL, FastAPI, Powershell ,Bash
- **Data Engineering:** Data Warehousing, ETL/ELT (SSIS), Apache Spark, Power BI , Azure Vms

January 2024 -July 2025 | Data Engineering Consultant | FNB | I was on Contract through Expeditus |

Worked as a Data Engineering Consultant (via Expeditus) and Business Intelligence Analyst II in the iDigiTech – I&SS team, focusing on Power BI platform management, BI infrastructure, and security. Managed Power BI workspaces, datasets, refresh schedules, gateways, and RLS, while building API watchers to monitor data pipeline and refresh failures. Designed and optimized ETL pipelines and Power BI data models to support self-service analytics and enterprise reporting, improving data reliability and supporting FNB's data-driven decision-making.

- **Programming:** Python, SQL,
- **Data Engineering:** ETL, Data Warehousing, Streaming, Pipeline Optimization
- **API Development:** RESTful APIs, GraphQL, System Integration
- **System Support:** Troubleshooting ETL failures

January 2024 -July 2025 | Data Engineering Consultant | Nedbank | I was on Contract through Expeditus |

I was contracted through Expeditus as a **Data Engineer at Nedbank**, specializing in financial risk and regulatory data systems. I designed and automated resilient data pipelines supporting interest rate, liquidity risk, and Funds Transfer Pricing (FTP) reporting, improving the accuracy and timeliness of daily risk and structural mismatch reports. I engineered SQL-driven ETL workflows and ingestion pipelines across Azure, Netezza, and Power BI, collaborating closely with finance, risk, and governance teams to meet regulatory and audit requirements. I supported Basel-related audits, resolved complex data issues via JIRA and ServiceNow, and contributed to disaster recovery initiatives, ensuring audit-ready, transparent, and resilient enterprise data infrastructure.

- **Cloud:** Microsoft Azure
- **Programming & Frameworks:** Python, SQL, NoSQL.
- **Version Control :** Github and Azure DevOps
- **Data Engineering:** SSIS, ETL/ELT, Data Warehousing, Real-Time Streaming, Pipeline Optimization, Power BI
- **Data Processing & Optimization:** Distributed Processing, Real-Time Analytics, Reporting, Performance Tuning
- **System Support:** Troubleshooting ETL failures

January 2022 - December 2023 | 2 Years | Data Science Internship | University of Pretoria | Contract

I did applied data science and AI research as a Computer Science Intern in the Department of Computer Science at the University of Pretoria (Tuks) for 2 years and they paid my fees in return , conducting data analysis and machine learning experiments using Python and SQL. I supported academic research activities through data preprocessing, feature engineering, and model development, and built Power BI dashboards and visualisations to effectively communicate financial data insights. I collaborated with researchers on projects involving South African metropolitan financial data analytics and AI-driven language recognition systems, and co-authored and published a research paper in the field of Artificial Intelligence [Paper Title “ Dipolelo: An automatically collected, prepared, and benchmarked dataset for the South African Sign language”](#).

Technical Skills & Tools

- **Version Control:** GitHub, Azure DevOps , Microsoft Azure
- **Databases:** PostgreSQL, SQL Server (Managed)
- **Development Tools:** Visual Studio Code
- **Operating Systems:** Windows Server, Linux
- **Programming & Scripting:** Python, SQL, PowerShell

January 2020 – December 2025 | Programming Instructor (Undergraduate Support) | University of Pretoria | Contract

I taught programming to undergraduate students on an ongoing, part-time basis, supporting their learning through hands-on instruction, practical labs, and guided problem-solving. My focus was on helping students understand core programming concepts, apply theory in practice, and develop the technical skills required for both academic coursework and real-world applications.

Skills & Technologies:

- **Programming Languages:** Python, SQL , PostgreSQL, SQL Server
- **Data Warehousing:** Data modeling, ETL fundamentals
- **Applied Data Science:** Data analysis, practical problem-solving
- **Version Control:** GitHub

Achievements :

- Top 15 Geeks in South Africa (2023) – Geekulcha
- Winner of 7 Hackathons, demonstrating innovation and problem-solving excellence
- Mentor at CSIR, NEMISA, and MTN, supporting skills development and large-scale coding initiatives

Skills :

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- Programming: Python, Golang, Ruby, SQL, Bash/Shell Scripting
 - Data Engineering: Apache Spark, Databricks, Snowflake, dbt, ETL/ELT Pipelines, SSIS
 - Cloud & Containers: Microsoft Azure , Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Azure DevOps)
 - Business Intelligence: Power BI, Data Modeling, DAX, SQL Analytics
 - Data & Analytics: Data Warehousing, Data Quality, Data Governance, Performance Optimization
 - Version Control & DevOps: Git, GitHub, GitLab, Infrastructure as Code (IaC)

References

Available upon request.